Knowledge Based Engineering (KBE) AE4204

Dr.ir. G. La Rocca

Flight Performance & Propulsion Group (FPP)

Intro

Learning objectives
Course organization
Assessment policy

Instructors & communication

Dr.ir. G. (Gianfranco) La Rocca

Course coordinator &
instructor (FPP)
Office 7.21

Dr.-Ing. A. (Alexander)
Heidebrecht

Instructor (FPP)

Office 7.10

- For personal questions or related to course organization: G.LaRocca@tudelft.nl
- For course content and (assignment) technical questions: tudAE4204@gmail.com

Contents of this intro lecture

- High level goals
- KBE – What are we talking about
- Learning objectives
- Set-up and organization
- Study material
- Assessment policy
- Detailed course agenda

- **High level Goals**
- **KBE taxonomy**
- **Learning objectives**

High level course goals

- Introduce **Knowledge Based Engineering (KBE)** to (aerospace) MSc students
- Prepare students for MSc thesis work involving:
  - the use and/or development of KBE applications
  - the advancement of KBE methodology

KBE – what are we talking about?

- **Knowledge-Based Engineering (KBE)** is a **methodology** that formalizes, captures, and reuses engineering knowledge **to automate and support design and analysis processes**

- The computational technologies used to implement KBE are typically provided by dedicated software platforms, referred to as **KBE systems** (often delivered as software development kits)

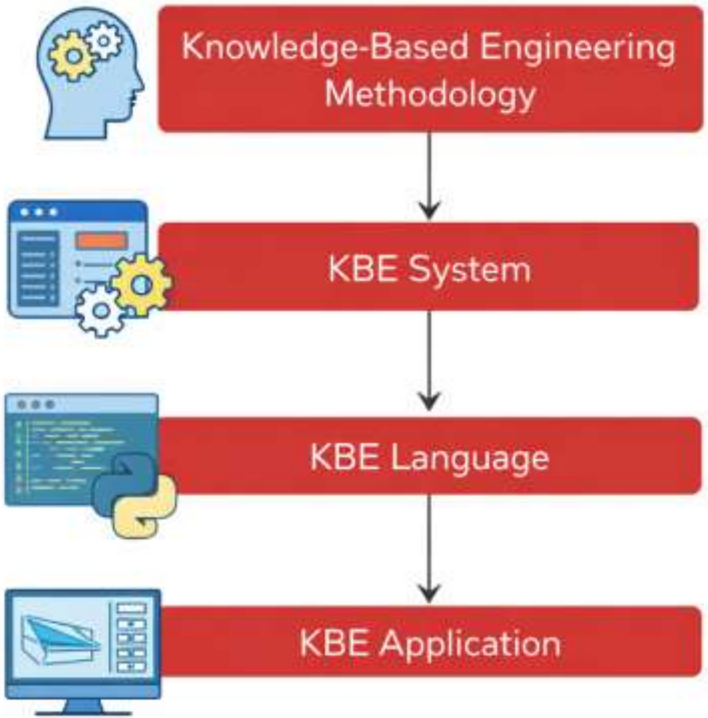
- A **KBE developer** uses a KBE system to build a **KBE application** by writing code in the programming language provided by that system

- ….

KBE – what are we talking about?

- ….
- The language provided by a KBE system includes specific constructs and mechanisms that support knowledge representation, reasoning, and automation, and is therefore referred to as a **KBE language**
- A **KBE application** is the resulting software artifact that uses the encoded engineering knowledge to automate and support a specific design and analysis task
- In this course, you will use a commercial KBE system, **ParaPy**, which provides a Python-based KBE language for developing KBE applications.

KBE – what are we talking about?



ParaPy, AML, GDL, …

Based on Python, based on Common Lisp,…

What you will develop for course assessment

Learning objectives

ParaPy

LO1 – KBE foundations and applicability

Apply the principles of Knowledge-Based Engineering (KBE) and evaluate when and why KBE is an appropriate approach to support the development of complex, multidisciplinary engineering products.

- Understand the role of KBE within modern digital engineering and product development workflows.
- Identify opportunities where KBE can improve decision-making and automation of analysis and design processes

Learning objectives

LO2 – Problem structuring and decision support

Design a logical and modular structure for a KBE application, based on *object-oriented modelling* principles, that represents a multi-component, multidisciplinary engineering problem and **supports engineering decision-making**.

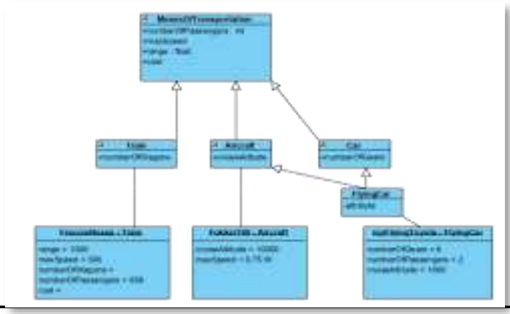
- Select appropriate **inputs, outputs, and functionality** to support relevant engineering decisions.
- Define a structure that links **engineering rules and external analysis methods** in a way that is consistent with the intended functionality.

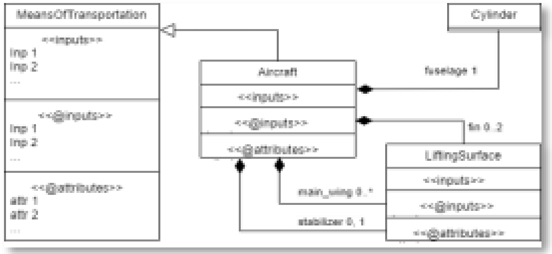
Learning objectives

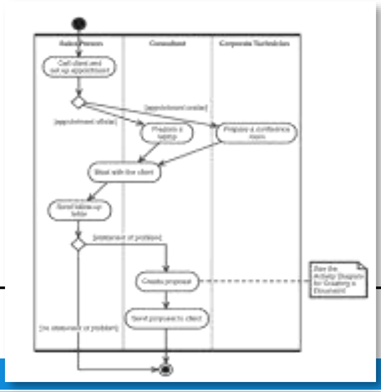
LO3 – Knowledge modeling and formalization

Formalize and communicate engineering knowledge using **UML** (Unified Modeling Language) **models** to describe the structure, behavior, and dependencies of a KBE application.

- Develop UML diagrams that reflect product architecture, design logic, and knowledge dependencies.
- Use models as a **design and communication tool**, not only as documentation.





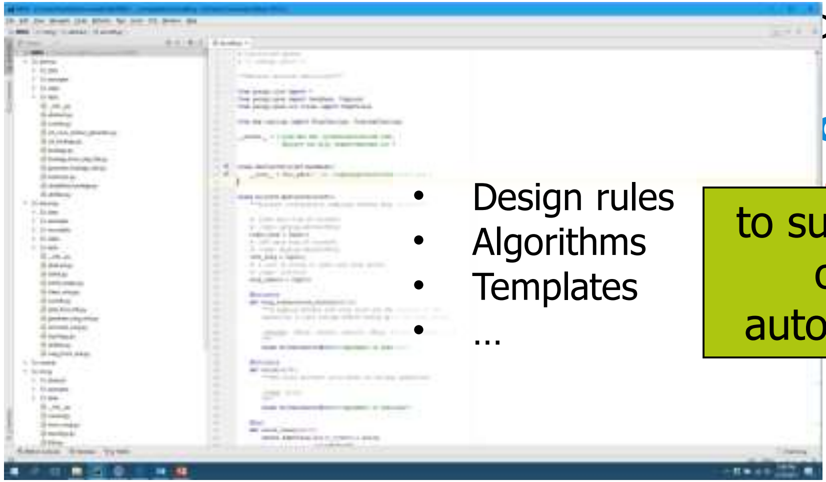


Learning objectives

ParaPy

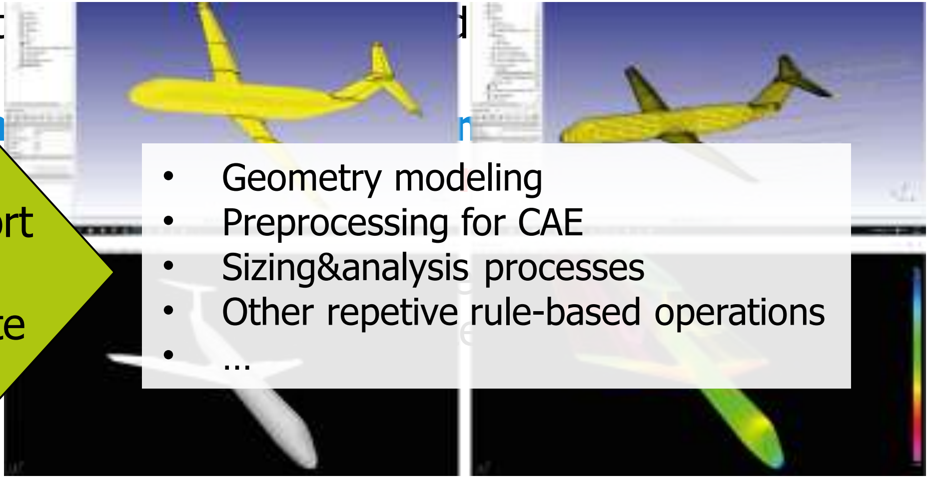
LO4 – KBE implementation

Implement a KBE application using a given KBE programming environment (ParaPy).



- Design rules
- Algorithms
- Templates
- …

to support or automate



- Geometry modeling
- Preprocessing for CAE
- Sizing&analysis processes
- Other repetive rule-based operations
- …

Course set up and organization

Course Set-up and Organization

Q3

- Theory lectures & demos
- Object-Oriented modeling tutorials with UML
- Computer-based tutorials on ParaPy KBE system (self-study& in-class discussion)
- Guest lectures

Q4

- KBE app development (self-study)
- Examinations
- Q&A/support sessions

(self-study) Python refresh

Self-study computer-based tutorials on ParaPy KBE system

How are your **Python programming** skills?

Start refreshing Python or learn the basics with one of the many online open courses*

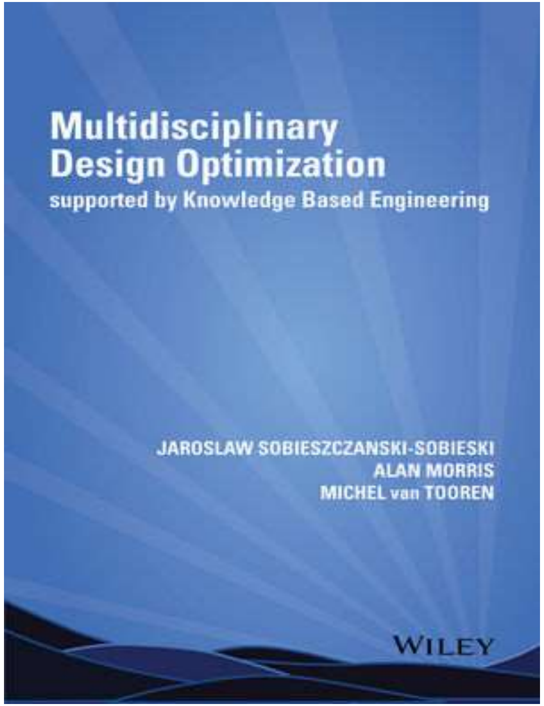
Study material

Course and Study Material

- Lecture sheets
- Some extra notes & interesting reading (journal papers, etc.)
- Software tool (Para*Py*) with documentation and tutorials (code samples)
- One suggested textbook

Refer to the sections *Lecture sheets* and *Extra notes & Interesting readings* on

Course and Study Material



Multidisciplinary Design Optimization Supported by ***Knowledge Based Engineering***

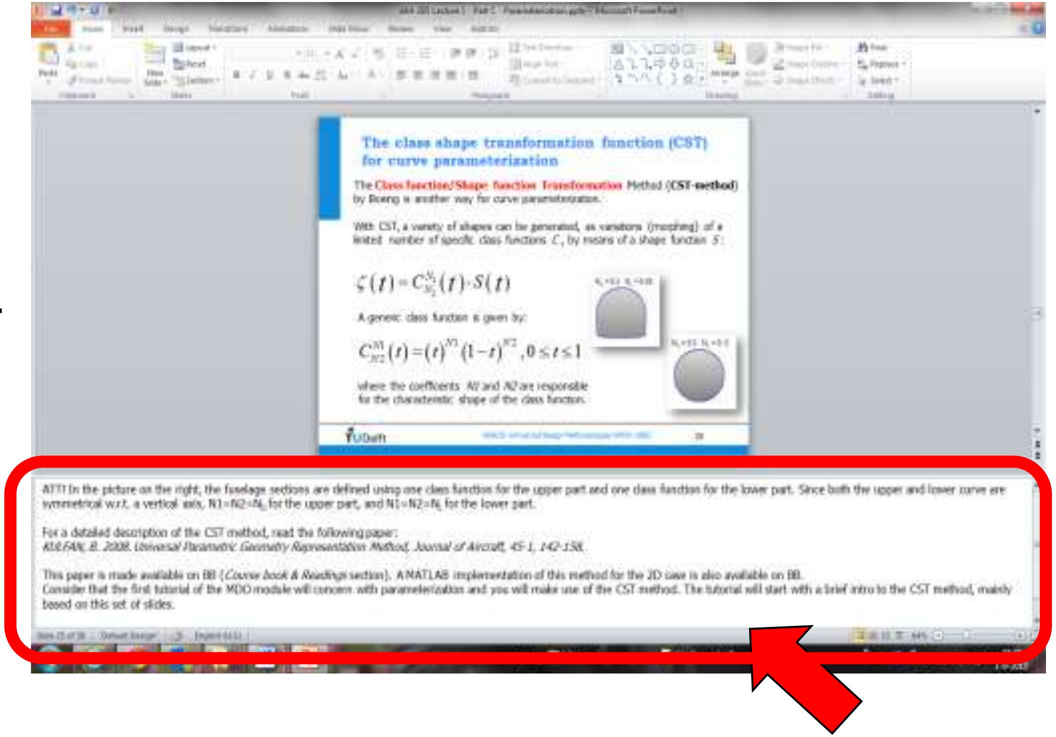
Jaroslaw Sobieszczanski-Sobieski,
Alan Morris,
Michel van Tooren
Gianfranco La Rocca contributor Ch.9 on KBE

ISBN: 978-1-118-49212-3

- You need **only CHAPTER 9 (pp 208-257)** for this course*
- You can download the PDF or access the whole e-book on the TU Delft library
- You can purchase hardcopy through **VSV Leonardo** for a discounted price

Notes…on the lecture notes

1. If a slide is hidden in presentation mode, it does not mean it is not interesting for you to read
2. Check the space below the slide. Typically, it is used for extra notes, examples, clarifications…
3. When you find an asterisk* on a word, it means a relevant note is provided in the space below the slide



Assessment policy

Assessment policy

- **ONE take-home assignment** to be performed **in teams of TWO students***
- Make your team by **signing-up** for one of the KBE Teams on Brightspace
- Deliverables: **KBE app & docs + oral exam** (live demo of your app + Q&A)
- **2 examination sessions** (nominal + retake) per academic year**.
- Team can be changed from one session to another
- In case of collaboration issues, any of the team members or instructors can split a team

Details in section *Assignment & grading policy & deadlines* on

Assessment policy

- You can make **your own** assignment proposal, or one based on a topic proposed **by a company**[1]
- Compulsory use of assignment **proposal template** [2]
- Assignment **proposal requires approval** by course instructors to access examination session **(pass/fail gate with iterations)**[3]
- For each examination session, a **separate assignment proposal must be delivered and approved**, according to the set deadlines
- **Reuse** of assignment proposal approved for the nominal session is allowed for retake session, under conditions:
    - **grade correction of -1 point** to account for the additional development time
    - **penalty-free reuse** if nominal examination grade >=5 AND <=7

Course agenda

Course calendar

2nd SEMESTER

Week no.	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
Week type	V	C	C	C	C	CT	C	C	CW	CWT	T	C	C	C	C	CT	C	C	CW	CWT	T
Teaching week	Spring break	3.1	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	3.10	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10
Monday	2	9	16	23	2	9	16	23	30	Easter Monday	13	20	King's Day	4		18	Whit Monday	1	8	15	22
Tuesday	3	L	L	I	L T	T	T	L G	31	7	14	21	28	Libera-tion Day	12	19	26	2	9	16	23
Wednesday	4	11	18	25	4	11	18	25	1		15	22	29	6	13		27	3	10		24
Thursday	5	12	19	26	5	12	19	26	2	9	16	23	30	7	Ascen-sion Day	21	28	4	11	18	25
Friday	6	L	L T	T	L	T	T	G	Good Friday	10	17	24	1	8	15	22	29	5	12	19	26
Saturday	7	14	21	28	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27
Sunday	8	15	22	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28

February | March | April | May | June

Lecture | Guest lecture
Tutorial | Installation

Assignment proposal Delivery deadline

KBE app delivery deadline

Examination sessions